Lecture 14 Worksheet - Vector Dialect and Tensorization

Name: ____________________   Date: ____________________

Exercise A - vector.transfer_read Annotation

```
%a = vector.transfer_read %A[%m, %k], %c0_i8
  {permutation_map = affine_map<(d0, d1) -> (d0, d1)>,
   in_bounds = [true, true]}
  : memref<?x?xi8, #sram>, vector<16x32xi8>
```

Field	Your annotation
base memref and memory space	
indices and tile origin	
vector shape and tile meaning	
permutation_map	
in_bounds / padding / mask policy	

Exercise B - vector.contract MatMul Mapping

```
#trait = {
  indexing_maps = [
    affine_map<(m, n, k) -> (m, k)>,
    affine_map<(m, n, k) -> (k, n)>,
    affine_map<(m, n, k) -> (m, n)>
  ],
  iterator_types = ["parallel", "parallel", "reduction"]
}
%c = vector.contract #trait %a, %b, %acc
  : vector<16x32xi8>, vector<32x16xi8> into vector<16x16xi32>
```

Concept	Answer
M dimension	
N dimension	
K/reduction dimension	
lhs layout	
rhs layout	
accumulator type	

Exercise C - NPU Tensorization Legality

Target native tile: M=16, N=16, K=32, lhs/rhs=i8, acc=i32, layouts A=MK, B=KN, full-tile only.

Case	Tensorize? Y/N	Reason	Fallback/Repair
vector<16x32xi8> x vector<32x16xi8> -> vector<16x16xi32>			
vector<16x48xi8> x vector<48x16xi8> ->			

vector<16x16xi32>			
vector<16x32xf16> x vector<32x16xf16> -> vector<16x16xf32>			
same shape, but rhs map is (n,k)			
same shape, but masked edge tile			

Exercise D - Rewrite to npu_kernel.matmul

아래 빈칸을 채워 target-specific pseudo IR 을 작성하세요.

```
%c2 = npu_kernel.matmul %a, %b, %acc
  {m = ____ : i64, n = ____ : i64, k = ____ : i64,
   lhs_layout = "____", rhs_layout = "____", acc_type = "____"}
  : vector<____x____xi8>, vector<____x____xi8>, vector<____x____xi32>
    -> vector<____x____xi32>
```

Exercise E - FileCheck Regression Test

```
// RUN: mlir-opt %s --npu-vector-contract-tensorize | FileCheck %s
// CHECK-LABEL: func.func @m16n16k32_i8
// CHECK: ____________________________
// CHECK-SAME: _______________________
// CHECK-NOT: ________________________
```

Reflection

tensorization 을 막아야 하는 조건 3 가지는?

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____

mask edge tile 을 어떻게 처리할 것인가?

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vector.transfer_read 의 permutation_map 이 layout bug 를 만드는 사례는?

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